Solve the following equations:

$$2^{x+1} + 2^{x-1} = 40$$

$$2^{x+1+x-1} = 40$$

$$2^{2x} = 40$$

$$2^{2x} = 2^2(10)$$

$$\log_2(40) = 2x$$

$$x = \frac{\log_2(40)}{2} \quad \Rightarrow \quad 40 = 2^3(5)$$

$$x = \frac{\log_2(2^3) + \log_2(5)}{2}$$

$$\boxed{x = \frac{3 + \log_2(5)}{2}}$$

$3^{2x} = 81$

We Know $81 = 3^4$

$3^{2x} = 3^4 \implies 2x = 4$

$x = 2$

$2^{3x} = 50$

$\log_2(50) = 3x$

$x = \dfrac{\log_2(50)}{3}$ $\quad \cancel{50 = 2^5 + (18) =}$